THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Geography Chapters 5 & 6: **Minerals and Energy Resources, Manufacturing Industries** | Class 10 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 2000	Few are abundant
2	**(a)** limestone	Most rocks have several
3	**(c)** peninsular rocks	Distribution
4	**(a)** Both A and R are true and R is the correct explanation of A	Unequal distribution
5	**(a)** 50–60 per cent	Most important industrial ore
6	**(b)** secondary sector	Manufacturing
7	**(a)** eradication of unemployment and poverty	Also reducing regional disparities
8	**(b)** SAIL	Government owned
9	**(a)** Both A and R are true and R is the correct explanation of A	Textile weakness
10	**(b)** Maharashtra, Gujarat and Tamil Nadu	Weaving decentralised

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Geographers study minerals as part of the earth's crust for landforms and their distribution and economic activities; geologists study the formation, age and composition of minerals.	2
12	Smaller occurrences of minerals in cracks, crevices, faults or joints of igneous and metamorphic rocks are veins; larger ones are lodes.	2
13	Increase in the value of raw materials by processing them into finished products of higher value, e.g. cotton to fabric.	2
14	All its raw materials as well as finished goods are heavy and bulky, entailing heavy transportation costs.	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Malleable, ductile and good conductor; used in electrical cables, electronics and chemical industries; India is critically deficient; Balaghat (MP), Khetri (Rajasthan), Singhbhum (Jharkhand).	3
16	Weaving supplies low-quality fabric as it cannot use much of the high-quality yarn produced in the country; traditional industries suffered during colonial rule against mill cloth from England; the industry must compete internationally; mill loomage is kept lower than powerloom and handloom to protect employment of weavers.	3
17	Growing energy consumption increases dependence on fossil fuels; rising prices and potential shortages of oil and gas threaten energy security; fossil fuels cause environmental problems; India has abundant sunlight, water, wind and biomass.	3
	OR: Domestic and industrial fuel; fuel in power sector; heating in industries; raw material in chemical, petrochemical and fertiliser industries; transport fuel (CNG); cooking fuel (PNG).	3
18	Sugarcane is harvested in a particular season, so mills crush only for some months; cooperatives pool resources of farmers, share profits, reduce exploitation and work efficiently.	1+2
	OR: Fast-growing and diversifying; inorganic chemicals for fertilisers, synthetic fibres, plastics, adhesives, paints, dyes, glass, soaps; organic chemicals for synthetic fibres, rubber, plastics, dyes, drugs and pharmaceuticals; both large and small units.	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Mica — plates/leaves, di-electric strength, electric and electronic industries; Koderma–Gaya–Hazaribagh (Jharkhand), Ajmer (Rajasthan), Nellore (AP). Limestone — calcium carbonates in sedimentary rocks; cement, iron smelting. Bauxite — aluminium; Amarkantak, Maikal hills, Bilaspur-Katni; Odisha (Panchpatmali).	5

	OR: Energy needed in all sectors; consumption rising steadily; India among least energy-efficient countries; fossil fuels limited and polluting; twin planks — energy conservation and renewable energy; citizen action — public transport, switching off lights, power-saving devices.	5
20	Backbone of development; modernises agriculture; jobs in secondary and tertiary; reduces poverty and unemployment; reduces regional disparities; exports and foreign exchange; agro-industries raise productivity; competitiveness in globalisation.	5
	OR: Cotton — raw cotton, market, transport, ports, labour, moist climate; Maharashtra, Gujarat; spinning in Maharashtra, Gujarat, TN. Jute — Hugli basin: jute areas, water transport, water for processing, cheap labour, Kolkata's facilities; WB.	5

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Geography Chapters 5 & 6: **Minerals and Energy Resources, Manufacturing Industries** | Class 10 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Jharkhand	Iron ore states
2	**(c)** the hump of an ox	Kudre = horse
3	**(b)** Marmagao port	Goa
4	**(c)** copper	Copper imports
5	**(a)** three-fourths	Base for metallurgy
6	**(b)** Bangladesh (erstwhile East Pakistan)	Mills remained in India
7	**(c)** Uttar Pradesh and Bihar	Sugar industry
8	**(c)** the index of a country's development	Basic industry
9	**(b)** Odisha	Also WB, Kerala, UP, Chhattisgarh
10	**(b)** glass, soaps, detergents and paper	Inorganic chemicals

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	It is the basic mineral for iron and steel, on which all other industries depend for machinery.	2
12	Assam; Digboi, Naharkatiya, Moran-Hugrijan (any two).	1+1
13	Public — BHEL, SAIL; private — TISCO, Bajaj Auto Ltd., Dabur Industries.	1+1
14	Manufacture of aircraft, utensils and wires; substitute for steel, copper, zinc and lead.	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	It has rich deposits of iron ore, manganese, coal, mica, bauxite, copper and uranium due to its ancient peninsular rock formations.	3
16	Liberalisation brought new and contemporary models; demand stimulated; passenger cars, two and three-wheelers; centres around Delhi, Gurugram, Mumbai, Pune, Chennai, Kolkata, Lucknow, Indore, Hyderabad, Jamshedpur, Bengaluru.	3
17	India has great wind potential; largest cluster from Nagarcoil to Madurai (TN); farms in AP, Karnataka, Gujarat, Kerala, Maharashtra, Lakshadweep; Nagarcoil and Jaisalmer known for effective use.	3
	OR: Hydro — from fast-flowing water, renewable, multi-purpose projects; thermal — from burning coal, petroleum, natural gas, non-renewable and polluting.	1½+1½
18	Fitting generators with silencers; redesigning machinery to reduce noise and increase energy efficiency; using noise-absorbing materials; ear plugs and earphones for workers.	3
	OR: Low cost of iron ore; high-grade raw materials in proximity (iron ore, coal, manganese, limestone); cheap labour; vast growth potential in the home market.	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Next to coal; fuel, lubricants, raw materials; nodal industry; anticlines and fault traps; Mumbai High, Gujarat (Ankleshwar), Assam (Digboi, Naharkatiya, Moran-Hugrijan); refineries; pipelines.	5
	OR: Basis of industry; iron and steel, aluminium, cement, electricals; energy minerals; exports of iron ore (Kudremukh, Bailadila); employment; challenges of mining hazards; conservation.	5
20	Cement — construction of houses, factories, bridges, roads, dams; bulky raw materials, coal, power, rail transport; first plant Chennai 1904; Gujarat plants for Gulf; Automobile — trucks, buses, cars, two and three-wheelers; growth after liberalisation; centres listed.	5

	OR: Air, water, land, noise pollution; Bhopal gas tragedy; effluents in rivers; fly ash; thermal pollution; remedies — effluent treatment, recycling water, rainwater harvesting, precipitators and scrubbers, cleaner fuels, silencers, energy-efficient machines, NTPC model.	5

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET C

Geography Chapters 5 & 6: **Minerals and Energy Resources, Manufacturing Industries** | Class 10 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** firewood and cattle dung cake	Conventional
2	**(a)** Both A and R are true and R is the correct explanation of A	Biogas alternative
3	**(a)** 200	Metallurgical coal
4	**(d)** Meghalaya, Assam, Arunachal Pradesh and Nagaland	About 55 million years old
5	**(c)** crest of the upfold	Porous layers
6	**(b)** oil refineries or petrochemical plants	Petrochemicals
7	**(a)** Both A and R are true and R is the correct explanation of A	Agro-linkage
8	**(d)** limestone, silica and gypsum	Bulky and heavy
9	**(b)** Shimla	Others: Delhi, Gurugram, Mumbai etc.
10	**(b)** toxic gas leak causing air pollution	Industrial hazard

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Heat and electricity produced by using the heat from the interior of the earth; hot groundwater rising as steam drives turbines.	2
12	It combines strength like iron with extreme lightness, good conductivity and great malleability.	2
13	Availability of raw cotton, market, transport including ports, labour and moist climate.	2
14	Uses latest techniques, minimises waste by ash utilisation, provides green belts, manages ash ponds and liquid waste, and does ecological monitoring.	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Unsafe narrow tunnels; risk to miners' lives; environmental degradation and water contamination; declared illegal by the National Green Tribunal.	3
16	Basic industries supply raw materials to other industries (iron and steel, copper and aluminium smelting); consumer industries produce goods for direct use by consumers (sugar, toothpaste, paper, sewing machines, fans).	3
17	High-grade hematite in Badampahar mines of Mayurbhanj and Kendujhar districts (Odisha); in adjoining Singhbhum district of Jharkhand, hematite mined in Gua and Noamundi.	3
	OR: Concentration of mineral in the ore; ease of extraction; closeness to the market; type of formation; cost of extraction.	3
18	Production of nitrogenous fertilizers (urea), phosphatic fertilizers, DAP and complex fertilizers (NPK); potash imported; expanded after Green Revolution; Gujarat, TN, UP, Punjab, Kerala produce half.	3
	OR: Thermal pollution of water by hot water discharge; fly ash; air pollution by burning coal; nuclear wastes cause cancers, birth defects and miscarriages.	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Nuclear, solar, wind, biogas, tidal, geothermal — locations and advantages; India's largest renewable programmes; reduces dependence on fossil fuels; environmental conservation; rural benefits.	5
	OR: Durg–Bastar–Chandrapur — Bailadila, super high grade hematite, exported to Japan and South Korea via Visakhapatnam; Ballari–Chitradurga–Chikkamagaluru–Tumakuru — Kudremukh, one of the largest deposits, 100 per cent export unit, slurry pipeline to Mangaluru.	5

20	Raw material (iron and steel near ores; sugar near cane), power (aluminium), labour (jute), market, transport (ports), water (jute processing), capital, banking and insurance (Kolkata), government policy (backward areas), climate (cotton mills in moist climate).	5
	OR: Raw materials — iron ore, coking coal, limestone (4:2:1), manganese; heavy industry; process — blast furnace, steel making, rolling; Chota Nagpur concentration; importance for all industries; index of development.	5

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ANSWER KEY & MARKING SCHEME SET D

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SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** Ankleshwar	Western India
2	**(b)** Krishna–Godavari basin	Natural gas
3	**(c)** 18,500 km	Gas Grid
4	**(d)** Chota Nagpur plateau	Koderma–Gaya–Hazaribagh
5	**(b)** Ajmer	Also Nellore belt in AP
6	**(b)** electrostatic precipitators	Also scrubbers
7	**(c)** screening, grinding, flocculation and sedimentation	Mechanical
8	**(d)** sugar industry in Maharashtra	Also coir in Kerala
9	**(b)** IT industry	Software
10	**(d)** noise pollution	Industrial noise

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	It is the basic raw material for the cement industry and essential for smelting iron ore in the blast furnace.	2
12	Saving energy through efficient use is equivalent to producing new energy, as it reduces demand on limited resources.	2
13	Undesirable gases like sulphur dioxide and carbon monoxide, and particulate matter like dust, sprays, mist and smoke.	2
14	In a globalised world, self-sufficiency is not enough; goods must match international quality to compete in world markets.	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Life processes cannot occur without minerals; though mineral intake is only about 0.3 per cent of nutrients, without them we cannot utilise the other 99.7 per cent of foodstuffs.	3
16	IT and electronics industry has generated large-scale employment, especially for educated youth; growth in hardware and software exports.	3
17	Constructed the 1,700 km HVJ pipeline linking Mumbai High and Bassein gas fields to fertiliser, power and industrial complexes in western and northern India; network expanded to 18,500 km and expected over 34,000 km as Gas Grid.	3
	OR: Electricity from oceanic tides; floodgate dams across inlets trap water at high tide; when the tide falls, water flows back to the sea through a pipe with a power-generating turbine; Gulf of Khambhat, Kuchchh, Sunderbans.	3
18	Primary — mechanical: screening, grinding, flocculation, sedimentation; secondary — biological processes; tertiary — biological, chemical and physical processes, including recycling of wastewater.	3
	OR: Industries set up in tribal and backward areas create jobs and infrastructure; public sector industries aimed at balanced regional development.	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Basic requirement for development; rising consumption; limited resources; pollution; ways — public transport, switch off lights, power-saving devices, non-conventional sources, efficient machinery and vehicles, awareness.	5
	OR: Pulmonary diseases from dust and fumes; mine roof collapse, flooding, fires; water contamination; land and soil degradation; river pollution; measures — strict safety regulations, environmental laws, protective equipment, reclamation, monitoring; avoid rat-hole mining.	5

20	ISO 14001 certification; optimum use of equipment with latest techniques; minimising waste by ash utilisation; green belts and afforestation; ash pond management, ash water recycling, liquid waste management; ecological monitoring and online database.	5
	OR: Definitions with BHEL/SAIL, TISCO/Bajaj/Dabur, OIL, Maharashtra sugar/Kerala coir; cooperative sector most suitable for sugar because it is seasonal, farmers supply raw material and share profits.	5

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